

Dr. Doc:

Welcome to pharmacology class

DRUGS FOR DISORDERS OF THE RESPIRATORY SYSTEM

Asthma Overview

Definition:

A chronic inflammatory disease.

Pathophysiology:

Characterized by

Airway inflammation,

Inflammation causes: Mucosal edema, Increased mucus production

Bronchial hyper-reactivity, and

Bronchospasm.

Major Symptoms of Asthma

Symptoms include

- shortness of breath,

- cough,

- chest tightness,

- wheezing, and

- rapid respiration.

- Classification of Asthma Severity

- Asthma is classified as Intermittent or Persistent (Mild, Moderate, Severe).

- Criteria for Severity:

- Includes frequency of symptoms,

- nighttime awakenings,

- frequency of SABA use for relief, interference with normal activity, and lung function tests (FEV1 and FEV1/FVC ratio).

- Treatment Goals and Drug Classes

- Goals of Therapy:

- Decrease intensity/frequency of symptoms,

- prevent future exacerbations, and

- minimize activity limitations.

- Drug Categories:

- Anti-inflammatory drugs and

- Bronchodilators.

- Administration:

Most treatments are administered via inhalation.

Classification of Bronchial Asthma Drugs

- Bronchodilators: Include

β₂ Sympathomimetics

Methylxanthines, and

Anticholinergics.

- Anti-inflammatory Agents: Include

Leukotriene(LT#) antagonists,

Mast cell stabilizers,

Corticosteroids (Systemic and Inhalational), and

Anti-IgE antibodies.

Bronchodilators Mechanism of Action

Stimulate β₂-receptors

β₂-receptors, increasing cAMP formation in bronchial muscle cells, which leads to muscle relaxation.

Note: These drugs provide symptomatic relief but have no anti-inflammatory effects.

Short-acting β₂ Agonists (SABAs)

Use:

✓ For quick relief of symptoms.

☐ Pharmacokinetics:

☐ Rapid onset (5 to 30 minutes) with relief lasting 4 to 6 hours.

☐ Indications:

✓ Appropriate for mild, intermittent asthma or exercise-induced bronchospasm (EIB).

☐ They should not be used as monotherapy for persistent asthma.

☐ Examples: Albuterol (Salbutamol), Levalbuterol, and Terbutaline.

☐ Long-acting β_2 Agonists (LABAs)

☐ Action:

✓ Provide bronchodilation for at least 12 hours and are taken on a fixed schedule.

☐ Requirement:

✓ Must only be used in combination with an asthma controller, such as an inhaled corticosteroid (ICS).

☐ Examples: Salmeterol and Formoterol

☐ Ultra-long-acting:

✓ Indacaterol, Olodaterol, Vilanterol, and Bambuterol (primarily for COPD).

☐ Adverse Effects of β_2 -agonists

☐ Side effects include tachycardia/angina

☐hyperglycemia,

☐hypokalemia, and

☐hypomagnesemia.

☐ Mitigation:

☐These adverse effects are minimized when the medication is inhaled.

☐Anti-cholinergic Drugs

☐ Use: Primarily used for COPD.

☐ Examples:

☐Ipratropium (short-acting, 4-6 hours) and

☐Tiotropium (long-acting, 24 hours).

☐ Mechanism:

✗block muscarinic cholinergic receptors in the bronchi to prevent bronchoconstriction and mucus secretion.

☐ Adverse Effects:

☐Xerostomia (dry mouth) and a bitter taste.

☐ Methylxanthines (Theophylline) Mechanism

☐ Mechanism of Action:

☐ Inhibition of phosphodiesterase (PDE), which increases cyclic nucleotide concentrations.

☐ Blockade of adenosine receptors.

☐ Release of Ca^{2+} from the sarcoplasmic reticulum.

☐ Result: Bronchodilatation in the bronchial muscle.

☐ Theophylline Pharmacokinetics

☐ Well absorbed orally and 50% protein-bound.

☐ Metabolism: Extensively metabolized in the liver (demethylation and oxidation) by CYP1A2.

☐ Half-life ($t_{1/2}$):

7–12 hours in adults, 3–5 hours in children, and slower in the elderly.

☐ Theophylline Safety and Interactions

☐ Safety:

Has a narrow margin of safety; side effects include headache, nervousness, nausea, cardiac arrhythmias, and seizures.

☐ Drug Interactions:

☐ Metabolism Enhancers:

Phenytoin, rifampicin, and phenobarbitone.

☒ Metabolism Inhibitors☒:

Erythromycin, ciprofloxacin, and cimetidine.

☢ Glucocorticoids Introduction and MoA

☒ Role:

Drugs of choice for long-term control of airway inflammation.

☒ Mechanism:

☒ Inhibit Phospholipase A2 to prevent the release of arachidonic acid, leading to direct anti-inflammatory effects.

☒ Additional Effects:

☒ Decrease the inflammatory cascade (eosinophils, macrophages, and leukocytes).

☒ Reduce airway mucosal edema and mucus production.

☒ Increase the number and responsiveness of bronchial β_2 receptors.

☒ Glucocorticoid Formulations and Status Asthmaticus

☒ Inhaled Glucocorticoids (IGC):

☐ Examples include

☐ beclomethasone, budesonide, fluticasone, mometasone, and triamcinolone.

☐ Oral Corticosteroids (CS):

Examples include methylprednisolone, prednisone, and prednisolone.

☐ Status Asthmaticus:

Severe exacerbations may require intravenous methylprednisolone or oral prednisone to reduce airway inflammation.

☐ Adverse Effects of Glucocorticoids

☐ Common Effects:

☐ Oropharyngeal candidiasis, ☐ dysphonia, and
☐ sore throat.

☐ Systemic Effects (Long-term/High dose):

☐ Adrenal suppression,

☐ slow growth in children/adolescents, and

☐ bone loss.

❓ Oral-Specific Effects:

❓Hyperglycemia and peptic ulcer disease.

❓ Pregnancy:

❓Inhaled steroids are considered safe during pregnancy.

❓ Mast Cell Stabilizers (Sodium Cromoglycate)

❓ Action:

❓Acts as a prophylactic anti-inflammatory agent by ❓inhibiting mast cell degranulation and the subsequent release of histamine, leukotrienes (LTs), and interleukins (ILs).

❓ Clarification:

❓It is not a bronchodilator and does not antagonize the constrictor actions of histamine, acetylcholine (Ach), or LTs.

❓Mast Cell Stabilizer Uses, PK, and AEs

❓ Therapeutic Uses:

❓Used as an alternative therapy, for exercise-induced bronchospasm (EIB), and allergic rhinitis.

❑ Pharmacokinetics (PK):

❑ It is not absorbed orally,

❑ available as a nebulized solution,

❑ has a short duration of action, and is rapidly excreted unchanged in urine and bile.

❑ Adverse Effects (AEs):

Includes cough, throat irritation, and an unpleasant taste.

❑ Leukotriene Modifiers Overview

❑ Zileuton:

❑ An inhibitor of 5-lipoxygenase (5-LOX) that prevents the formation of both LTB₄ and cysteinyl leukotrienes.

❑ Zafirlukast and Montelukast:

Selective antagonists of the cysteinyl leukotriene-1 (cysLT₁) receptor.

❑ Usage:

☐ They are alternatives to inhaled glucocorticoids for mild-to-moderate asthma and can have an additive effect in severe asthma.

☐ They are effective for aspirin-induced and exercise-induced asthma.

Leukotriene Synthesis Pathway

Pharmacokinetics of Leukotriene Modifiers

☐ Metabolism:

They are orally active, highly protein-bound, and undergo extensive hepatic metabolism.

- ☐ Half-life ($t_{1/2}$): Montelukast (3–6 hours), Zafirlukast (8–12 hours), and Zileuton (2.5 hours).

☐ Excretion:

Zileuton is excreted in urine; Zafirlukast and montelukast undergo biliary excretion.

☐ Interaction: Food reduces the absorption of zafirlukast.

Adverse Effects and Interactions of Leukotriene Modifiers

☐ Neuropsychiatric Effects:

These include depression, anxiety, abnormal dreams, hallucinations, insomnia, and suicidal thinking.

❓ Specific Drug Effects:

Zafirlukast can cause headache and GI disturbance. Zileuton is hepatotoxic and inhibits CYP1A2.

❓ Enzyme Inhibition:

Zafirlukast inhibits CYP2C8, 2C9, and 3A4.

Monoclonal Antibodies

❓ Omalizumab:

✔ Selectively binds to human IgE, decreasing its binding to receptors on mast cells and basophils; it is reserved for resistant, allergic asthma.

❓ IL-5 Antagonists:

Mepolizumab, benralizumab, and reslizumab are interleukin-5 antagonists.

Monoclonal Antibodies Indications and Limitations

❓ Indication:

Treatment of severe persistent asthma poorly controlled by conventional therapy.

❑ Limitations:

❑ High cost, specific administration routes (IV for reslizumab, SC for others), and a significant adverse effect profile.

❑ Adverse Effects of Monoclonal Antibodies

❑ Side effects include

rare anaphylactic reactions, arthralgias, fever, rash, and an increased risk of infections.

Pharmacotherapy of Cough

❑ Definition:

❑ Cough is a protective reflex intended to expel secretions and foreign particles.

❑ Classification:

Can be productive or non-productive; identifying the cause is essential before treatment.

Expectorants and Mucolytics

❑ Expectorants (e.g., Guaifenesin):

❑ Increase bronchial secretions or reduce viscosity to facilitate removal by coughing.

☐ Mucolytics (e.g., Hypertonic saline, Acetylcysteine):

☐ React directly with mucus to make it less viscous.

Antitussives and Codeine

☐ General:

☐ Antitussives suppress cough centrally and peripherally and should only be used for dry, non-productive coughs.

☐ Codeine:

⚠ Decreases the sensitivity of the CNS cough center.

☐ At high doses, it can cause respiratory depression and drowsiness (especially in children); AEs include constipation, fatigue, and addiction.

☐ Pholcodine

☐ Properties: Similar antitussive efficacy to codeine but has practically no analgesic or addicting properties.

☐ Duration: It is longer-acting, with effects lasting up to 12 hours.

☐ Opioid Antitussives - Pholcodine

❓ Properties:

❓ It has practically no analgesic or addicting properties.

❓ Efficacy:

Its antitussive efficacy is similar to that of codeine.

❓ Duration:

It is longer-acting, with effects lasting up to 12 hours.

❓ Nonopioid Antitussives - Dextromethorphan

❓ Mechanism:

It is a synthetic central N-methyl D-aspartate (NMDA) receptor antagonist.

❓ Key Characteristics:

❓ It does not depress the mucociliary function of the airway mucosa and is practically free of constipating effects.

❓ Adverse Effects:

☒ Common effects include dizziness, nausea, and drowsiness; high doses may result in hallucinations and ataxia.

☒ Adjuvant Antitussives

☒ Antihistamines:

✓ Often added to antitussive or expectorant formulations.

☒ First-generation antihistamines are commonly used, including Chlorpheniramine, Diphenhydramine, and Promethazine.

☒ Bronchodilators: Examples include Salbutamol and Terbutaline.

☒ Drugs Used to Treat Allergic Rhinitis

☒ Definition: Allergic rhinitis is an inflammatory disorder affecting the upper airway.

☒ Symptoms:

Includes sneezing, rhinorrhea (runny nose), pruritus (itching), and nasal congestion (caused by the dilation and increased permeability of nasal blood vessels).

☒ Forms: It exists in two major forms: seasonal and perennial.

☒ Antihistamines for Allergic Rhinitis

☐ Treatment Overview: Options include antihistamines, corticosteroids, and α -adrenergic agonists.

☐ H1 Receptor Antagonists: Second-generation antihistamines are primarily used, such as fexofenadine, loratadine, desloratadine, and cetirizine.

☐ Topical Intranasal Antihistamines: Includes olopatadine and azelastine, which provide increased drug delivery with fewer adverse effects.

☐ Corticosteroids for Allergic Rhinitis

☐ Intranasal Formulations: Examples include beclomethasone, budesonide, fluticasone, ciclesonide, mometasone, and triamcinolone.

☐ Onset of Action:

Ranges from 3 to 36 hours; however, for chronic rhinitis, improvement may not be seen for 1 to 2 weeks after starting therapy.

☐ Adverse Effects:

Nasal irritation, nose bleeds, sore throat, and rarely, candidiasis.

☐ α -Adrenergic Agonists (Nasal Decongestants)

☐ Mechanism:

☐ They activate α_1 -adrenoceptors in the nasal mucosa arterioles, causing vasoconstriction, which reduces edema and airway resistance.

☐ Examples: Phenylephrine, pseudoephedrine, and oxymetazoline.

☐ Usage Limitations:

Intranasal formulations should not be used for more than 3 days due to the risk of rebound nasal congestion.

☐ Oral vs. Intranasal:

☐ Oral formulations provide a longer duration of action but also cause increased systemic effects, such as elevated blood pressure, increased heart rate, excitation, and the potential for abuse (specifically pseudoephedrine).

☐ Theophylline is a methylxanthine derivative (chemically related to caffeine) used primarily as a bronchodilator in chronic asthma and COPD.

✦ Its mechanism of action is multifactorial, involving three distinct cellular targets:

☐ The Core Mechanisms of Action (The "Big Three")

Mechanism 1: Non-Selective ☐ Phosphodiesterase (PDE) ☐ Inhibition (Primary Bronchodilator Pathway)

☐ Target:

☐ Inhibits PDE3 and PDE4 enzymes.

Biochemical Cascade:

1. PDE normally $\rightarrow$ degrades intracellular cyclic AMP (cAMP) into 5'-AMP.
2. By blocking PDE, theophylline causes an accumulation of cAMP inside bronchial smooth muscle cells and inflammatory cells.
3. Increased cAMP activates Protein Kinase A (PKA).
4. PKA phosphorylates and $\rightarrow$ inactivates Myosin Light Chain Kinase (MLCK) while promoting calcium efflux out of the cytoplasm.

$\rightarrow$ Result:

$\rightarrow$ Bronchial smooth muscle relaxation (bronchodilation) and decreased release of inflammatory mediators from mast cells and eosinophils.

Mechanism 2: Competitive Adenosine Receptor Antagonism (The "Caffeine-Like" Pathway)

$\rightarrow$ Target:

$\rightarrow$ Blocks $\rightarrow$ cell-surface Adenosine (A₁ and A₂) receptors.

Physiological Effect:

$\rightarrow$ In asthmatic patients, endogenous adenosine causes smooth muscle bronchoconstriction and triggers mast cells to release histamine and leukotrienes.

By blocking adenosine receptors, theophylline prevents adenosine-induced bronchospasm.

Side Effect Link

Adenosine normally acts as an inhibitory neurotransmitter in the CNS and slows cardiac conduction.

Blocking adenosine receptors explains theophylline's classic toxic side effects:

CNS stimulation (seizures/insomnia) and

cardiac stimulation (tachycardia/arrhythmias).

Mechanism 3:

Activation of Histone Deacetylase 2 (HDAC2) (Anti-Inflammatory Pathway)

Target:

Recruits and activates HDAC2 in inflammatory cell nuclei.

Biochemical Cascade:

1. Inflammatory cytokines cause acetylation of histones, keeping DNA open for transcription of more inflammatory genes.

2. Theophylline turns on HDAC2, which deacetylates histones, causing DNA to wrap tightly.

3. This represses the transcription of pro-inflammatory genes (e.g., IL-8, TNF- α).